AN INVERTIBLE MODULE IS THE SAME THING AS A FINITE LOCALLY FREE MODULE OF RANK ONE, AND ITS INVERSE IS ITS DUAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes '(details omitted)' once, at proof line 36133, for the verification that the evaluation map becomes an isomorphism after localizing; that phrase is kept verbatim and the step is immediately closed by the subsequent cover argument.

Allowed standard prerequisites (editorial).

- Finite locally free modules and the cover argument for checking an isomorphism locally. used at proof lines 36131-36134 and 36162-36163. Lets the evaluation map be verified after localizing, and characterises finite local freeness.
- Invertible modules and tensor functors forming a quasi-inverse pair. condition (2) at statement line 36122 and used at proof lines 36138-36141. Is the equivalence with the existence of an inverse module.
- Automorphisms of a module factoring through a finite free module. used at proof lines 36148-36161. Produces the direct summand presentation of M that yields finite local freeness.

Source reference. Original lemma 00EO: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 542–542. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 00NX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 875–875. Cross-chapter navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let R be a ring. Let M be an R-module. Equivalent are*

(1) *M is finite locally free module of rank 1,*
(2) *M is invertible, and*

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(3) *there exists an R-module N such that $M \otimes_R N \cong R$.*

Moreover, in this case the module N in (3) is isomorphic to $\mathrm{Hom}_R(M, R)$.

Proof. Assume (1). Consider the module $N = \mathrm{Hom}_R(M, R)$ and the evaluation map $M \otimes_R N = M \otimes_R \mathrm{Hom}_R(M, R) \to R$. If $f \in R$ such that $M_f \cong R_f$, then the evaluation map becomes an isomorphism after localization at f (details omitted). Thus we see the evaluation map is an isomorphism by Algebra, Lemma 00EO. Thus (1) $\Rightarrow$ (3).

Assume (3). Then the functor $K \mapsto K \otimes_R N$ is a quasi-inverse to the functor $K \mapsto K \otimes_R M$. Thus (3) $\Rightarrow$ (2). Conversely, if (2) holds, then $K \mapsto K \otimes_R M$ is essentially surjective and we see that (3) holds.

Assume the equivalent conditions (2) and (3) hold. Denote $\psi : M \otimes_R N \to R$ the isomorphism from (3). Choose an element $\xi = \sum_{i=1,\ldots,n} x_i \otimes y_i$ such that $\psi(\xi) = 1$. Consider the isomorphisms

$$M \to M \otimes_R M \otimes_R N \to M$$

where the first arrow sends x to $\sum x_i \otimes x \otimes y_i$ and the second arrow sends $x \otimes x' \otimes y$ to $\psi(x' \otimes y)x$. We conclude that $x \mapsto \sum \psi(x \otimes y_i)x_i$ is an automorphism of M. This automorphism factors as

$$M \to R^{\oplus n} \to M$$

where the first arrow is given by $x \mapsto (\psi(x \otimes y_1), \ldots, \psi(x \otimes y_n))$ and the second arrow by $(a_1, \ldots, a_n) \mapsto \sum a_i x_i$. In this way we conclude that M is a direct summand of a finite free R-module. This means that M is finite locally free (Algebra, Lemma 00NX). Since the same is true for N by symmetry and since $M \otimes_R N \cong R$, we see that M and N both have to have rank 1. $\square$